FAKHUR U DIN

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RESEARCH DIRECTION / INTEREST

My work centers on practical, applied Machine Learning — building systems that automate the data-to-decision pipeline (preprocessing, model training, prediction, and explainability), as well as recommendation systems. I am interested in extending this into graduate research on AutoML and explainable AI (XAI).

ACADEMIC QUALIFICATION

Bachelor of Science in Information Technology (BS-IT) Jan 2023 – Expected Jan 2027
Shaheed Benazir Bhutto University, Shaheed Benazirabad (SBBU-SBA) Field of Study: Information Technology

RESEARCH & PROJECTS

Intelligent Data Analysis System Using AI Agents *(Final Year Project)*

The Intelligent Data Analysis System using AI Agents is a web-based platform that automates the complete data analysis process, from preprocessing to prediction. It allows users to upload datasets and automatically performs data cleaning, EDA, and machine learning. With AutoML and explainable AI, the system converts raw data into meaningful insights through a simple one-click pipeline.

- Automated Preprocessing: handles missing values, duplicates, outliers, and data types
- Smart EDA: generates summaries, trends, and correlation analysis
- AutoML: detects problem type, trains multiple models, and selects the best one
- Model Explainability: feature importance and confusion matrix for transparent results
- Interactive Dashboard: real-time results, data export, and automated reporting

Interactive Data Visualization Web App

Built an interactive Data Visualization App using Python, Streamlit, and Seaborn. Users can upload datasets, explore insights, and create dynamic plots (scatter, bar, histogram, box, pair, facet grid). The app features real-time customization, session-state logic, and a modern CSS-styled UI. Focused on clean, maintainable code, responsive design, and smooth user experience.

Tools: Python, Streamlit, Seaborn, pandas, CSS

AI-Powered Book Recommendation System

Developed an intelligent book recommendation system using Python. Implemented KNN with cosine similarity and Truncated SVD to improve accuracy and performance. Designed an interactive Streamlit web app that suggests similar books based on a user's selected title with appealing visual output.

Tools: Python, Scikit-Learn, pandas, NumPy, Streamlit

ACADEMIC PROJECTS

Tailor Management System, Real Estate Agency Management System, Text Editor, and File Sorter — desktop applications built with Python and Tkinter as part of coursework.

TECHNICAL SKILLS

Languages & Tools: Python, Tkinter **Data & ML Libraries:** pandas, NumPy, Scikit-Learn, Seaborn **Frameworks:** Streamlit

EXPERIENCE

May 2025 – July 2025

Artificial Intelligence Course Instructor (Part-time) @ *Muntaha Skills Development Center, Nawabshah*

- Delivered a batch-level course on AI/ML fundamentals, covering core concepts and practical applications
- Guided students through hands-on exercises using Python-based tools
- Evaluated student progress and provided individual feedback
- Simplified complex ML concepts for learners with limited technical background

CERTIFICATIONS & ACHIEVEMENTS

September 22 – November 16, 2025 **| People's Information Technology Programme (PITP) Phase-II, Government of Sindh**

Hands-on Training in Python Development

- *Completed a two-month hands-on program at Shaheed Benazirabad – SBBUSBA, certified by Sukkur IBA University; covered practical Python programming exercises and applied development skills.*

August 2025 **|2nd Position, Tech Exhibition Competition @ Shaheed Benazir Bhutto University, Shaheed Benazirabad**

- Secured 2nd position in a university-wide project exhibition competition, presenting the Interactive Data Visualization Web App against entries from IT-related departments during Pakistan Independence Day Celebrations.

REFERENCES

Dr. Mairaj Nabi | Associate Professor / Head of Department | Email: chairman.it@sbbusba.edu.pk | Phone: +92 3337016377
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